

National University of Singapore
 School of Computing
 CS1101S: Programming Methodology
 Semester I, 2026/2027
S5-in-class
Recursive Llist Processing

Problems:

1. The function `llist_ref` can be applied to a list `xs` and a number `n`, and returns the `n`-th element of the list, starting counting at 0. So `llist_ref(llist(1, 2, 3), 2)` evaluates to 3. The position of an element in the list is called its *rank*; we say that the number 3 has rank 2 in the list. Write a Python function called `every_second` that takes a list as its only argument and returns a list containing all the elements of odd rank (i.e. every second element) from the input list.

```
every_second(llist("a", "x", "b", "y", "c", "z", "d"))
# Value: ["x", ["y", ["z", None]]]
```

```
def every_second(items):
    ...
```

2. Write a Python §2 function called `sums` that takes a **llist of numbers** as its only argument and returns a **llist of numbers**, containing two elements: the first is the sum of all even-ranked numbers in the input llist (ranks 0, 2, 4 etc), whereas the second element is the sum of all odd-ranked elements in the input llist (ranks 1, 3, 5 etc).

Example call:

```
sums(llist(1, 2, 3, 4, 5))
# Value: [9, [6, None]]
```

We say that `sums` has the following type:

`sums`: llist of Number $\rightarrow$ llist of Number